



AI TRIPLE CAMERA
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EDC

Chapters-

1

1. a. What do you mean by dynamic resistance of a diode? Derive an expression for dynamic resistance of a diode.

Sol: Dynamic resistance of a diode is defined as the ratio of change in voltage to change in current.

$$\text{i.e. } r_d = \frac{dV_D}{dI_D}$$

we know,

$$I_D = I_S \left(e^{\frac{V}{\eta V_T}} - 1 \right)$$
$$\Rightarrow I_D = I_S e^{\frac{V}{\eta V_T}} - I_S \quad \text{--- (1)}$$

Now, differentiating (1) w.r.t. V , we get;

$$\Rightarrow \frac{dI_D}{dV} = \frac{I_S}{\eta V_T} e^{\frac{V}{\eta V_T}} - \frac{dI_S}{dV}$$

$$\Rightarrow \frac{dI_D}{dV} = \frac{I_S}{\eta V_T} e^{\frac{V}{\eta V_T}}$$

$$\Rightarrow \frac{dI_D}{dV} = \frac{(I_D + I_S)}{\eta V_T} \quad (\because \text{from (1)})$$

Since $I_D \gg I_S$, so I_S is neglected.

$$\frac{dI_D}{dV_D} = \frac{I_D}{\eta V_T} \quad (\because V = V_D)$$

$$\Rightarrow \frac{dV_D}{dI_D} = \frac{\eta V_T}{I_D}$$

$$\therefore r_d = \frac{\eta V_T}{I_D}$$

This is required expression for dynamic resistance of a diode.

where, $\eta = \text{constant } 1 \leq \eta \leq 2$

$V_T = \text{Thermal voltage}$

$I_D = \text{diode current}$

1. a. Explain the operation of PN diode in case of forward biased and reversed biased. Differentiate between Zener diode breakdown and avalanche breakdown.

1. Forward bias:

When a positive terminal of a source is connected to positive terminal of a battery (P-type) and the negative terminal of battery is connected to N-type

1.a. Explain the operation of PN diode in case of forward and reverse biased. Differentiate between zener breakdown and avalanche breakdown.

sq :

1) forward biased:

When a positive terminal of a battery is connected to P-type and the negative terminal of a battery connected to N-type, then the diode is called forward biased.

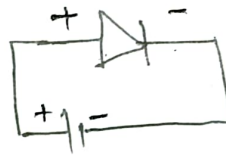
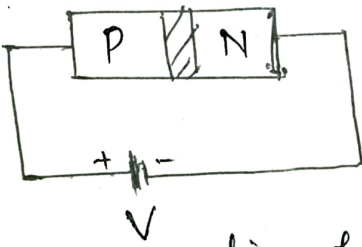


fig. forward biased.

In forward biased, the positive terminal of source repels the holes whereas the negative terminal of source repels the electrons due to which there is flow of holes and electron across the junction. Thus diode conduct current in forward biasing.

In forward biased, diode conducts current only if the supply voltage is greater than barrier potential of a diode.

~~In~~ When a diode is forward biased, the depletion region becomes narrower.

2) Reverse biased:

When a positive terminal of a battery connects is connected to n-type and the negative terminal of a battery is connected to p-type, then the biased is formed is ~~forward biased~~ reverse biased.

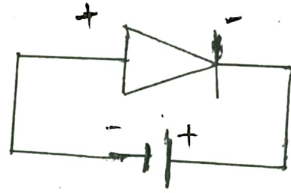
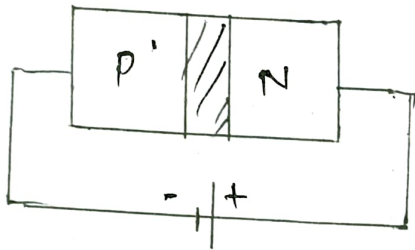


fig. ~~forward~~ biased
reverse

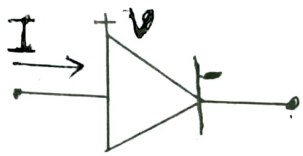
In reverse biased, the positive terminal of a source attracts electrons and the negative terminal of a source attracts holes, away from the junction. Due to the movement of electrons and holes the depletion region becomes wider and does not conduct current.

When a reverse biased voltage is greater than a breakdown voltage, the diode breakdown and acts as a s.c. so it conducts high current.

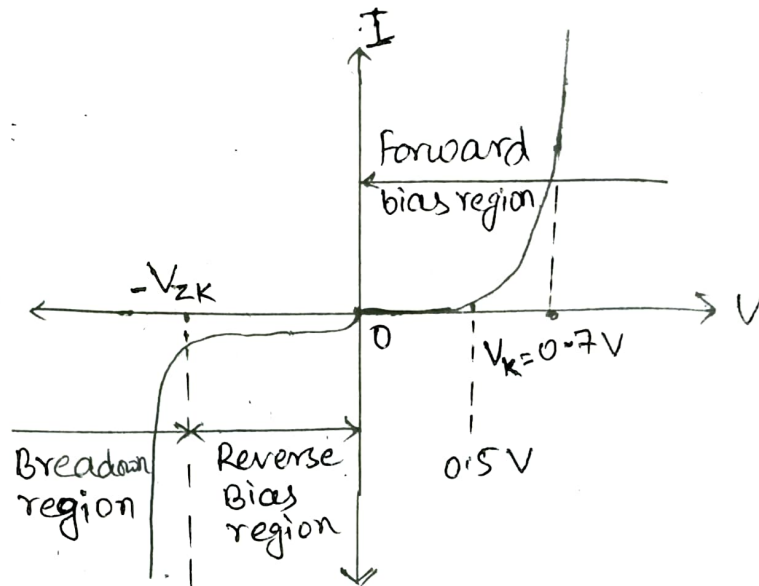
④ Differences between zener breakdown and avalanche breakdown.

S.N	Zener breakdown	S.N	Avalanche breakdown
1.	Zener breakdown is the process in which the electrons move across the barrier from the valance band of p-type material to conduction band of n-type material.	1.	Avalanche breakdown is the process of applying high voltage and increasing the free electrons in semiconductor and insulating materials.
2.	Zener breakdown is observed in zener diode having breakdown voltage 5 to 8 volts.	2.	Avalanche breakdown is observed in zener diode having breakdown voltage greater than 8 volts.
3.	In Zener breakdown, breakdown voltage decreases on increasing temperature.	3.	In Avalanche breakdown, the breakdown voltage increases on increasing temperature.
4.	It occurs in diode that are highly doped diode.	4.	It occurs in diode that are lightly doped diode.
5.	The VI characteristics of a zener diode has a sharp curve.	5.	The VI characteristics of a Avalanche breakdown has not as sharp curve as Zener diode.

- 1.a. What is V-I characteristics of diode? Explain the effects of temperature in V-I characteristics curve.
- solⁿ V-I characteristics of diode is the voltage and current relationship across the diode when it is in operation.



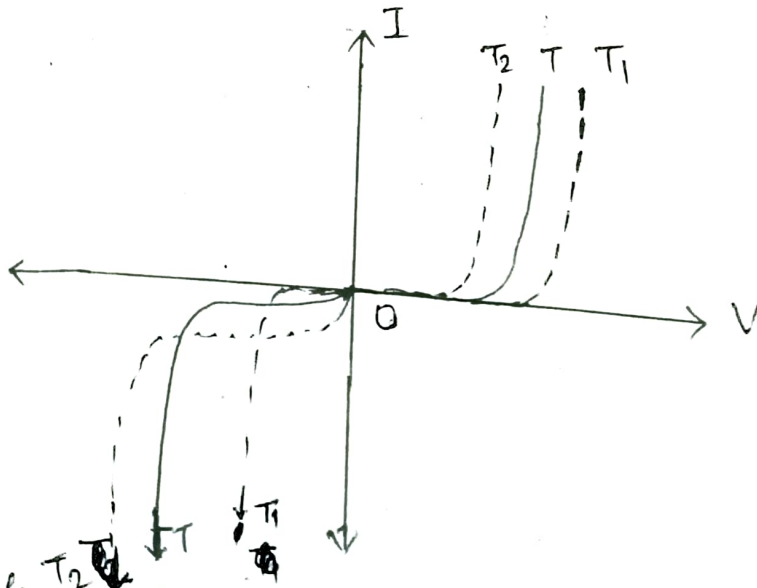
fig(a) diode with terminal current and voltage



fig(b) V-I characteristics curve of Si diode.

Effects of temperature on V-I characteristics curve:

Let us consider three different temperatures T , T_1 and T_2 such that $T_2 > T > T_1$ as shown in V-I characteristics curve.



The relationship between the current I and voltage V is given by

$$I = I_s \left(e^{\frac{V}{\eta V_T}} - 1 \right) \quad \text{--- (1)}$$

where I = diode current

I_s = reverse saturation current

V = diode voltage

η = emission coefficient, $1 \leq \eta \leq 2$

$V_T = \frac{KT}{q}$, thermal voltage

K = Boltzmann's constant = $1.38 \times 10^{-23} \text{ JK}^{-1}$

T = absolute temperature

q = electronic charge = $1.6 \times 10^{-19} \text{ C}$.

We observed that:

- i) As the temperature is increased, the curve in the forward direction shifts towards the origin. This means the forward voltage drop decreases with the increase in temperature. Hence, the diode conducts at even lower value of voltage forward voltage.
- ii) The reverse current increases with increase in temperature.

1. b. Describe Zener voltage regulation for all possible cases
sq To use zener diode as a voltage regulation, following conditions must be satisfied:

- i> Zener diode must be operated in reverse biased and connected in parallel with load.
- ii> Input voltage must be greater than zener voltage ($V_{in} > V_z$) i.e. it must be operated in breakdown region.

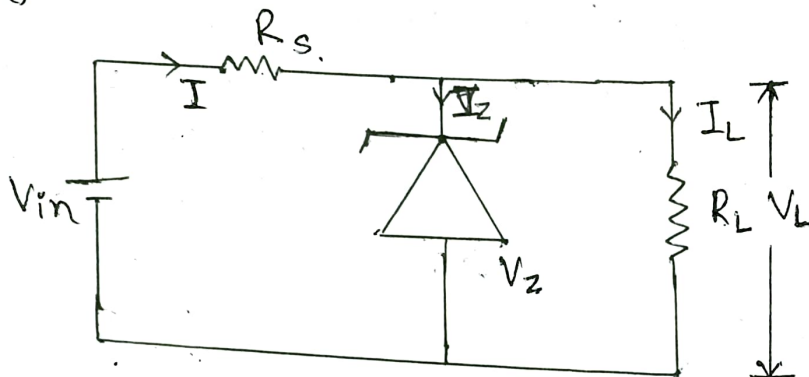


fig. Zener diode as a voltage regulator

where, V_{in} = input voltage
 R_s = source resistance
 I = source current
 I_z = zener current
 V_z = zener voltage
 I_L = load current
 R_L = load resistance
 V_L = load voltage

From figure,

$$I = I_z + I_L$$

$$\text{and } V_L = I_L R_L$$

$$V_{in} = I R_s$$

When zener diode regulates voltage across load, the load voltage is equal to zener voltage.
i.e. $V_L = V_Z$

$$\text{i.e. } V_Z = I_L R_L \quad \text{But } I_Z \neq I_L$$

case I: When V_{in} changes:

If V_{in} increases, the current I also increases. That is I_Z and I_L both increase but zener diode conducts I_Z maintaining I_L constant. Thus, $V_L = I_L R_L$ becomes constant when V_{in} increases.

case II: When R_L changes:

When R_L increases, the current I_L decreases and I_Z increases then the zener diode conducts more current. Thus $V_L = I_L R_L$ remains constant ~~on increasing~~ when R_L ~~changes~~ increases.

Definition:

Zener diode is a special diode designed to use as a voltage regulator.

1.a. Define Fermi-level in an intrinsic semiconductor, hence prove that "the Fermi level lies midway between the conduction and valence bands."

eg. The maximum energy level that an electron can get at the absolute zero temperature is known as Fermi level.

In an intrinsic semiconductor, the Fermi-level is defined as the midway between the conduction and valence band.

(*) For an intrinsic semiconductor, the energy band is:

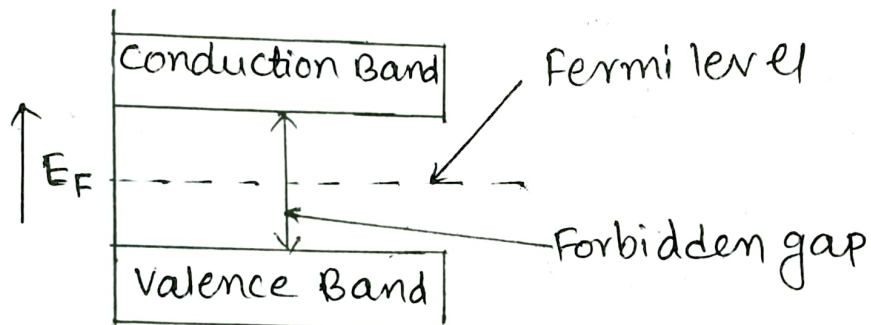


fig. Energy band

Fermi level on the middle of forbidden band indicates the hole concentration of the in valence band is equal to the electron concentration in conduction band.

$$\text{i.e. } p = n \quad \text{--- (1)}$$

The hole concentration in ~~conduction~~ valence band is given by

$$p = N_v e^{-\frac{(E_F - E_v)}{KT}} \quad \text{--- (2)}$$

where, N_v = effective density of states in valence band

K = Boltzmann's constant

E_F = Fermi level energy

E_v = Energy in valence Band

T = absolute temperature.

The electron ~~conduction~~ concentration in conduction band is given by $-(E_c - E_F)/KT$

$$n = N_c e^{-(E_c - E_F)/KT} \quad \text{--- (2)}$$

where, N_c = effective density of states in conduction band

E_c = Energy in conduction band

From (1), (2) and (3), we get:

$$p = n$$
$$\Rightarrow N_v e^{-(E_F - E_v)/KT} = N_c e^{-(E_c - E_F)/KT}$$

$$\Rightarrow \frac{N_v}{N_c} = \frac{e^{-(E_F - E_v)/KT}}{e^{-(E_c - E_F)/KT}}$$

$$\Rightarrow \frac{N_v}{N_c} = e^{(-E_F + E_v + E_c - E_F)/KT}$$

⇒ Taking log on both sides, we get:

$$\log\left(\frac{N_v}{N_c}\right) = \frac{(-2E_F + E_v + E_c)}{KT}$$

$$\Rightarrow KT \log\left(\frac{N_v}{N_c}\right) = E_v + E_c - 2E_F$$

$$\Rightarrow 2E_F = E_v + E_c - KT \log\left(\frac{N_v}{N_c}\right)$$

At absolute zero temperature, $T = 0K$.

$$2E_F = E_v + E_c$$

$$\therefore E_F = \frac{E_v + E_c}{2}$$

This proved that the fermi level lies mid way between the conduction band and valence band.

- 1.b. Discuss the formation of volt-Amp ($V-I$) characteristics of PN Junction. What happens when a P-N junction is biased in
- Forward direction.
 - Reverse direction.

~~The voltage an~~
The $V-I$ characteristics of a PN diode is the voltage and current relationship across the diode when it is in operation.

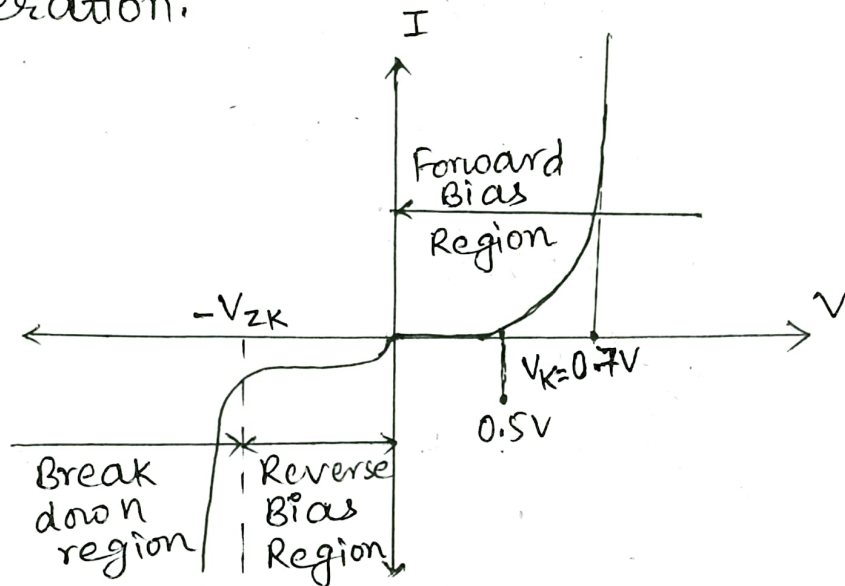


fig. $V-I$ characteristics of PN diode

⊗ Forward Bias Region:

→ When a diode is forward biased, small amount of current flows through it until the supply voltage is greater than barrier potential.

→ When the supply voltage exceeds the barrier potential, then current increases sharply as shown in figure.

⊗ Reverse Bias Region:

→ When a diode is reverse biased, negligible amount of current flows through it until the supply of voltage is greater than breakdown voltage.

→ When the supply exceeds the breakdown voltage, the diode is breakdown and acts as short circuit then the current increases sharply as shown in figure.

When a diode (PN junction) is forward biased:

When a positive terminal of a battery is connected to P-type and a negative terminal of a battery is connected to ~~n-type~~ n-type of a diode, then diode formed is called forward biased.

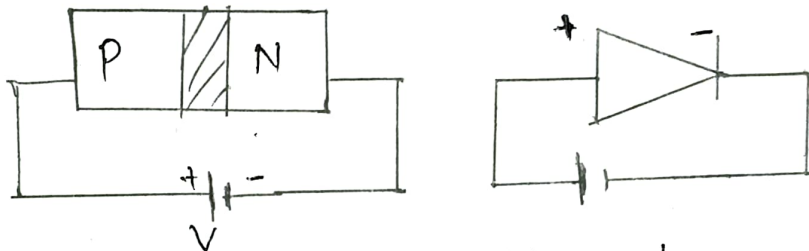


fig. Forward biased

In forward biasing, the positive terminal of a battery repels the holes whereas the negative terminal of a battery repels electrons due to which there is flow of electrons and holes across the junction. Thus diode conducts current in forward biasing.

In forward bias, diode ~~current~~ conducts currents only if the supply voltage is greater than barrier potential of a diode.

~~The~~ When a diode is forward bias, the depletion region becomes narrower.

27 When a P-N junction is reverse biased:

When a positive terminal of a battery is connected to n-type and negative terminal of a battery is connected to p-type of a diode, then the reverse bias is formed.

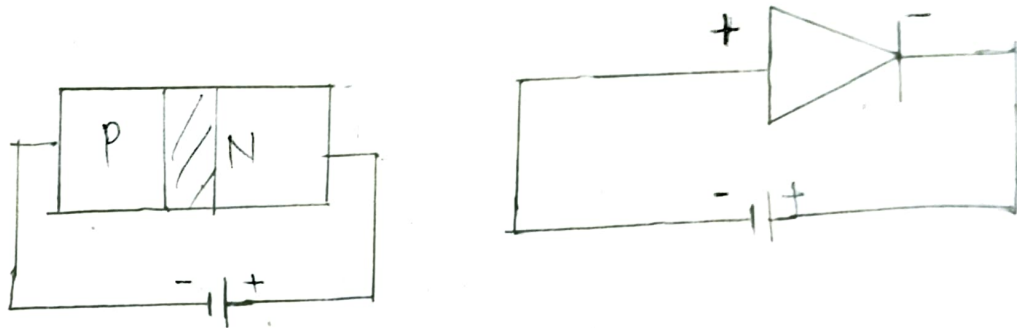


Fig. Reverse biased

In reverse biasing, the positive terminal of a battery is connected to attracts electrons and the negative terminal of a battery attracts holes from semi-conductor diode, away from the junction. Due to movement of electrons and holes, the depletion region widens and diode doesnot conduct current.

When a reverse biased voltage is greater than a breakdown voltage, the diode is breakdown and acts as short circuit and then conducts high current.

~~1. a show that $R_{DP} = R^2$ for darlington pair~~

1.a. Differences between Diffusion Capacitance and Transition Capacitance: "Transition capacitance is inversely proportional to the square root of the reverse voltage applied across it." Verify this statement.

sof Differences between Diffusion Capacitance and Transition Capacitance:

S.N	Diffusion Capacitance	S.N	Transition Capacitance
1.	The capacitive effect shown by diode when it is forward biased is called diffusion Capacitance.	1.	The capacitive effect shown by diode when it is reverse biased is called transition capacitance.
2.	The diffusion capacitance is directly proportional to the forward current of diode.	2.	The transition capacitance is inversely proportional to the reverse voltage of diode.
3.	The diffusion Capacitance of a diode is expressed as $C_D = \frac{I\tau}{\eta V_T}$	3.	The transition Capacitance of a diode is expressed as $C_T = \frac{\epsilon A}{W}$

Transition capacitance is the capacitive effect shown by diode when it is in reversed biased.

The transition capacitance of a diode is given by

$$C_T = \frac{\epsilon A}{W} \text{ ————— (1)}$$

where, ϵ = permittivity

$A =$ ~~are~~ cross sectional area of a junction.
 $w =$ width of depletion region

The depletion width (w) is expressed by

$$w^2 = \frac{2\varepsilon V}{e} \left[\frac{1}{N_A} + \frac{1}{N_D} \right]$$

$$\Rightarrow w^2 = \frac{2\varepsilon V}{e} \left[\frac{N_A + N_D}{N_A N_D} \right]$$

$$\therefore w = \sqrt{\frac{2\varepsilon V}{e} \left(\frac{N_A + N_D}{N_A N_D} \right)} \quad \text{--- (2)}$$

where, $N_A =$ No. of acceptor atoms

$N_D =$ No. of donor atoms

$\varepsilon =$ permittivity

$e =$ electronic charge

$V =$ reverse voltage ($V = V_0 - V_R$) applied

~~$V_0 =$ Contact potential~~

~~$V_R =$~~

From (1) and (2),

$$C_T = \frac{\varepsilon A}{\sqrt{\frac{2\varepsilon V}{e} \left(\frac{N_A + N_D}{N_A N_D} \right)}}$$

$$\Rightarrow C_T = \varepsilon A \times \sqrt{\frac{e N_A N_D}{2\varepsilon V (N_A + N_D)}}$$

$$\text{i.e. } C_T = \frac{1}{\sqrt{V}}$$

This proves that the transition capacitance is inversely proportional to the square root of reverse voltage applied.

1.a. What is semiconductor material? Explain the importance of semiconductor material on electronic device with examples.

^{So?} ~~≠~~ Semiconductors are materials which have a conductivity between conductors and insulators.

The resistivity of a semiconductor material lies in the range of 10^{-5} to $10^6 \Omega m$.

Semiconductors can be pure elements such as ~~doping~~ silicon, germanium or compounds such as gallium arsenide or cadmium selenide.

Properties of semiconductors:-

- i> Semiconductor acts as insulator at zero kelvin. On increasing temperature, it works as a conductor.
- ii> Lesser power losses
- iii> Semiconductors are smaller in size and possess less weight.
- iv> Resistivity is higher than conductor and lesser than insulators.
- v> Resistance of semiconductor decrease with increase in temperature and vice-versa.

Semiconductors are used in almost all electronic devices. Without them, our life would be much different.

Their reliability, compactness, low cost and controlled conduction of electricity make them ideal to be used for various purposes in a wide range of components and devices. Transistor, diodes, photosensors, microcontrollers, integrated chips and much more are made up of semiconductors.

Some importance of semiconductors are:-

- i> They are highly portable due to the smaller size
- ii> They requires less input power.
- iii> Semiconductor devices are shockproof.
- iv> They have a longer lifespan.
- v> They are noise-free while operating.

Sta



1.a. What happens when a forward biased diode is suddenly reverse biased? Explain with necessary diagram.

sol. When a forward biased diode is suddenly reverse biased, the diode does not change its state from conduction to non-conduction.

The diode takes certain time to change its state from conduction to non-conduction called reverse recovery time of diode.

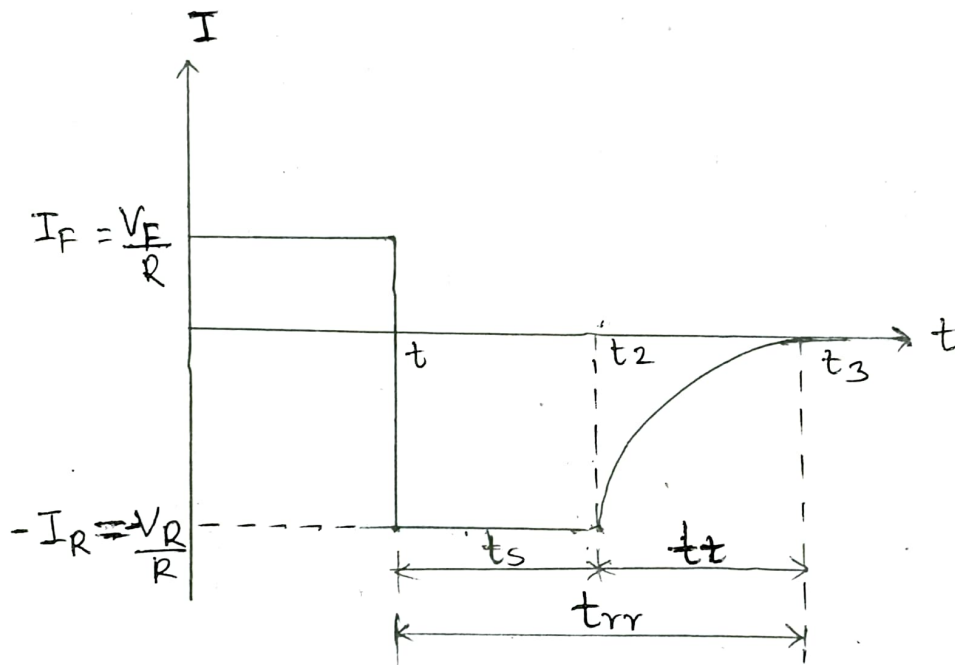


fig. reverse recovery time of a diode

When a forward biased diode is reversed, the current simply reverse and stay measurable for certain period, called storage time (t_s).

After storage phase, the current gradually decreases and becomes zero to take diode into non-conduction, state called transition time (t_t).

The reverse recovery time of a diode is the sum of storage time and transition time.

$$\text{i.e. } t_r = t_s + t_t$$

1.a. Differences between intrinsic semiconductor and extrinsic semiconductor.

Q: Differences between intrinsic and extrinsic s/c:

SN.	Intrinsic Semiconductor	SN.	Extrinsic Semiconductor
1.	Semiconductors without doping (i.e. pure).	1.	Semiconductors doping with impurities (i.e. impure)
2.	Conductivity is low.	2.	Conductivity is high.
3.	Number of free electrons and holes are equal. i.e. $n_e = n_h$	3.	Number of free electrons and holes are not equal. i.e. $n_e \neq n_h$
4.	Conductivity depends only on temperature.	4.	Conductivity depends on both temperature and impurity added.
5.	It does not conduct at zero Kelvin.	5.	It conducts at zero Kelvin.
6.	Conductivity is due to both holes and electrons.	6.	Conductivity is due to mainly due to majority charge carriers.



1.a. Draw an ideal and piecewise linear model of a diode.

sol

1. Ideal diode mode:

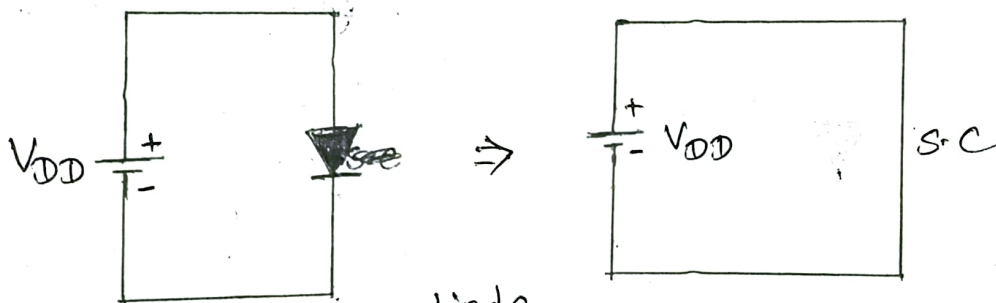
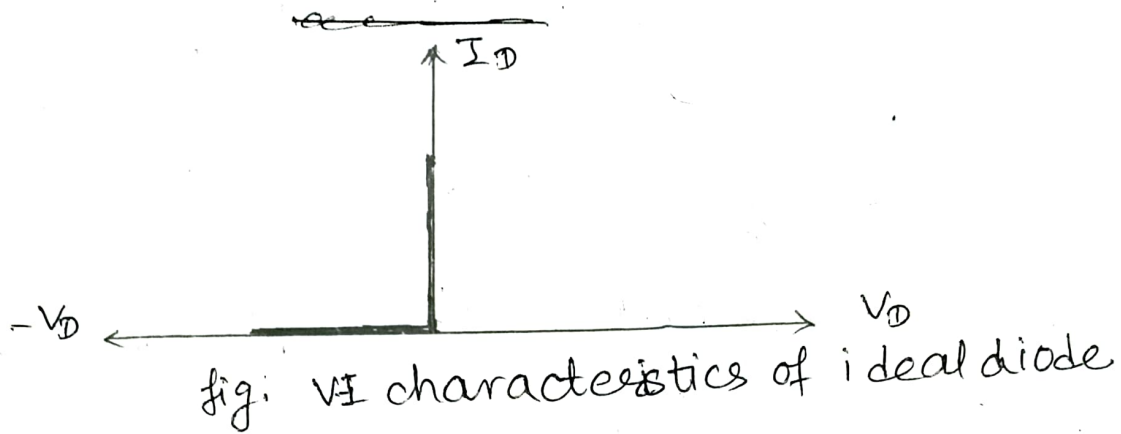


fig. Forward biased ^{diode} and ~~reverse~~ its equivalent circuit

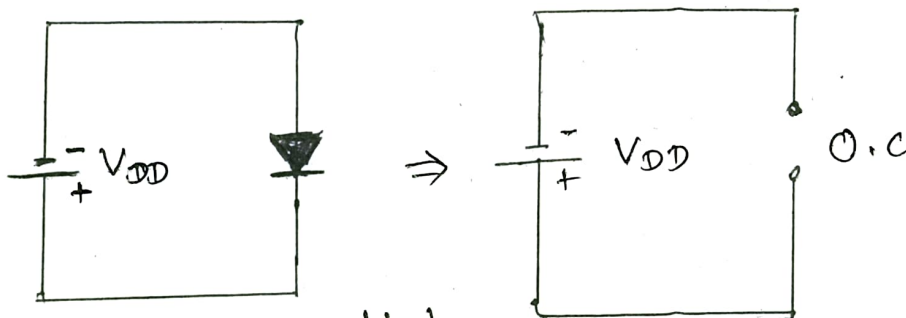


fig. Reverse bias ^{diode} and its equivalent circuit.

* Piecewise linear model of a diode:

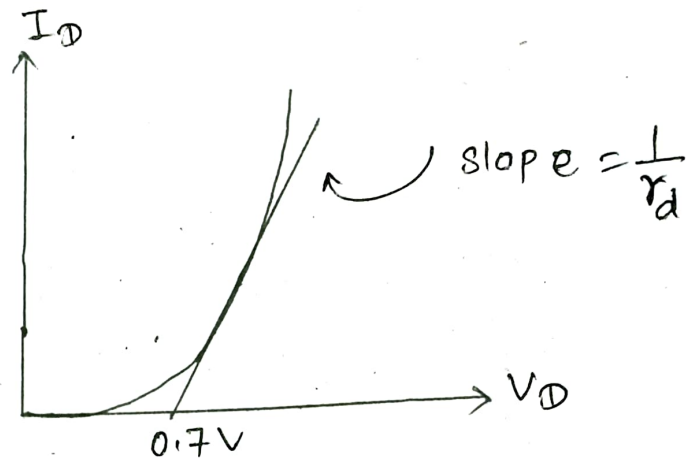


fig. V-I characteristics of piecewise model of a diode

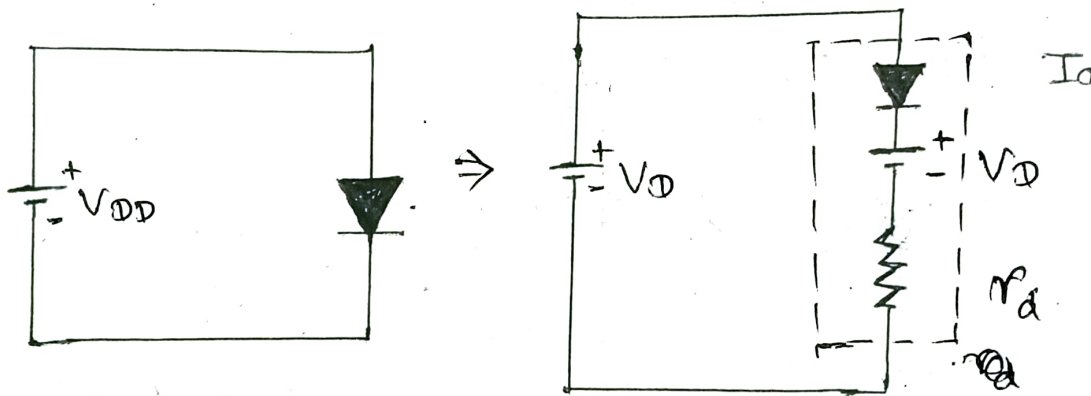


fig. piecewise model and its of diode and its equivalent circuit